

Invited Mini Review

Systems memory consolidation during sleep: oscillations, neuromodulators, and synaptic remodeling

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Memory consolidation transforms newly acquired experiences into stable long-term memories essential for learning and cognition. This process involves systems consolidation, where memory traces are reorganized across brain regions, and synaptic consolidation, which fine-tunes local neural connections. Sleep plays a critical role in both, coordinating memory reactivation, synaptic remodeling, and long-range neural communication. Systems consolidation is supported by stage-specific brain oscillations: during NREM sleep, the coupling of slow-oscillations, spindles, and sharp-wave ripples facilitates hippocampal-cortical transfer of memory representations, while REM sleep theta oscillations contribute to memory integration, abstraction, and emotional tagging. Complementary neuromodulatory dynamics, particularly involving norepinephrine and dopamine, regulate the timing and prioritization of memory processing. At the synaptic level, sleep balances strengthening and weakening of connections through a coordinated interplay of NREM and REM activity. Recent findings also suggest that dreaming may reflect the subjective correlate of these processes, particularly through the integration of recent and remote memory fragments. Although the precise relationship between systems-level reorganization and local synaptic refinement remains unclear—partly due to current technical limitations—emerging approaches are beginning to bridge these scales. Together, these findings underscore the integrative role of sleep in optimizing memory consolidation and offer promising avenues for clinical and translational research. [BMB Reports 2025; 58(10): 425-436]

INTRODUCTION

Memory consolidation is a fundamental process by means of

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which newly acquired, initially fragile, and easily modifiable short-term memories undergo stabilization, to ultimately transform into long-term memories that can persist for days, weeks, or even months (1, 2). This transformation is essential to learning and cognitive development, and ensures that relevant information is retained and integrated into existing knowledge structures. Memory consolidation is in general categorized into two distinct but interrelated mechanisms: systems consolidation, and synaptic consolidation. Systems consolidation refers to large-scale neural reorganization, and involves changes in synaptic connectivity and neural representations across distributed brain regions. In contrast, synaptic consolidation consists of a more localized process that strengthens synaptic connections within specific neural circuits, primarily at the level of individual neurons and synapses. Together, these processes contribute to the gradual stabilization and optimization of memory traces over time.

Among the various factors that influence memory consolidation, sleep plays a particularly critical role. Many studies have consistently demonstrated that sleep deprivation following learning experiences severely disrupts memory consolidation, leading to impaired retention and recall performance (3-5). These findings emphasize the importance of sleep-dependent mechanisms in stabilizing and restructuring memories. The prevailing theory of active systems consolidation suggests that during sleep, newly acquired memories—which are initially stored in the hippocampus—are gradually transferred and integrated into cortical networks, where they become more permanent, and resistant to interference. This interaction between the hippocampus and cortex is thought to be driven by coordinated neural activity that includes hippocampal sharp-wave ripples, cortical slow-oscillations, and thalamocortical spindles, all of which facilitate memory reactivation and redistribution.

Traditionally, active systems consolidation was considered most relevant to hippocampus-dependent declarative memories, such as episodic and spatial memories. However, recent research has challenged this distinction, revealing that even memories previously thought to be hippocampus-independent, such as procedural or skill-based motor memories, also involve hippocampal activity, particularly during the early stages of consolidation (6, 7). These findings suggest that the hippocampus plays a broader role in processing various types of memory beyond its classical domain.

In addition to increasing the long-term stability of memories, the consolidation process is associated with higher-order cognitive functions that include abstraction, generalization, and optimization (8). Through repeated reactivation and refinement, memories become more structured and interconnected, which enhances their accessibility and utility in novel situations. Moreover, as memories gradually shift from the hippocampus to distributed cortical regions, they contribute to the formation of knowledge networks that facilitate problem-solving, decision-making, and adaptive behavior. Understanding these mechanisms is critical to advance our knowledge of learning and memory, with potential applications in neuromodulation, neurorehabilitation, and artificial intelligence systems inspired by biological cognition.

Sleep stages and brain oscillations

Sleep is broadly categorized into rapid eye movement (REM)

sleep, and non-REM (NREM) sleep (Fig. 1A) (9). In human, NREM sleep is traditionally divided into four stages, whereas in rodent, sleep stages are shorter, and more fragmented. These sleep stages are primarily classified based on electroencephalogram (EEG) or local field potential (LFP) recordings, which capture the characteristic brain oscillations associated with each stage. During NREM sleep, distinct brain oscillations, particularly slow-oscillations, spindles, and sharp-wave ripples, are commonly observed, while REM sleep is primarily dominated by theta waves (Example recordings in the motor cortex and the hippocampus of a rat are shown in Fig. 1E).

Slow-oscillation: The first key oscillatory feature of NREM sleep is the slow-oscillation (Fig. 1B), a low-frequency wave of 0.1–4 Hz that is primarily observed in the cerebral cortex (9, 10). Slow-oscillations emerge due to alternating phases of synchronized neuronal activity: up-states, where large populations of neurons depolarize and fire together; and down-states, where neurons undergo hyperpolarization, and fall silent. These oscillations are thought to facilitate network synchronization, and play a critical role in memory consolidation.

Spindle: The second characteristic oscillation, the sleep spindle (Fig. 1B), is a transient burst of activity lasting 0.5–2 s, oscillating primarily at 10–15 Hz. Spindles originate in the thalamocortical system, particularly from GABAergic neurons in the *nucleus reticularis* of the thalamus, and are then transmitted to the cerebral cortex via thalamocortical pathways (10–12). These oscillations are associated with sensory gating to prevent external stimuli from disrupting sleep; they are also implicated in synaptic plasticity and memory processing (13).

Sharp-wave ripple: The third critical oscillatory pattern is the sharp-wave ripple (Fig. 1B), a high-frequency oscillation in the hippocampus that ranges 150–250 Hz. Sharp-wave ripples occur when sharp-waves generated in the CA3 region propagate to pyramidal neurons in CA1, leading to synchronous neuronal firing (14). Sharp-wave ripples play a fundamental role in memory replay and reactivation, where during sleep, neural representations of experiences from wakefulness are temporally compressed and reactivated (15–17). This replay mechanism is considered an important process that underlies memory consolidation (18, 19).

Theta wave: Lastly, during REM sleep, theta waves at 5–8 Hz (Fig. 1C) are predominantly observed in the hippocampus. These theta oscillations are driven by glutamatergic, GABAergic, and cholinergic neurons projecting from the medial septum to the hippocampus (20). Theta activity during REM sleep is considered critical for memory integration, emotional processing, and cognitive flexibility, and plays a particularly important role in spatial learning and navigation (21, 22).

Overall, these distinct brain oscillations across sleep stages are essential to consolidate memory and reorganize neural networks, and highlight the intricate relationship between sleep and learning.

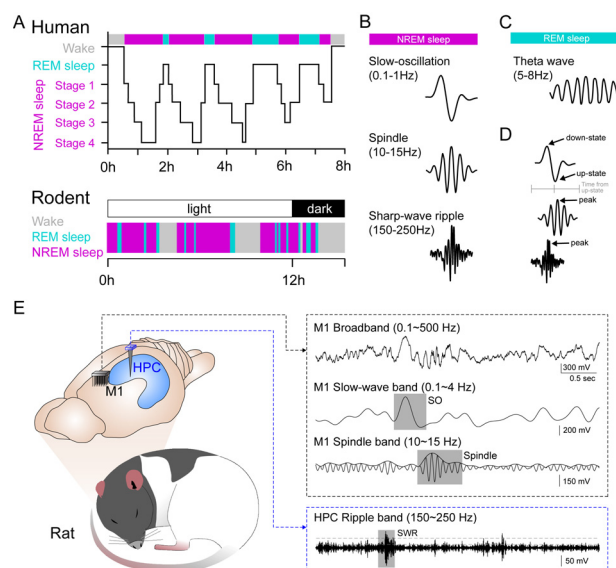


Fig. 1. Sleep architecture and neural oscillations. (A) Sleep/wake architecture of human and rodent. In human, NREM sleep is typically divided into four distinct stages, while in rodent, sleep stages are shorter, and more fragmented. These stages are primarily identified through electroencephalogram (EEG) or local field potential (LFP) recordings; these capture the characteristic brain oscillations associated with each sleep phase. (B) During NREM sleep, the key oscillatory patterns that are observed include slow-oscillations, spindles, and sharp-wave ripples. (C) During REM sleep, the dominant brain wave that is observed is the theta wave. (D) Timing and interaction of sleep oscillations. The triple coupling of slow-oscillation, spindle, and sharp-wave ripple is thought to facilitate both the transfer of reactivated hippocampal memories to the neocortex, and the synaptic consolidation necessary for memory reorganization. (E) Example of sleep LFP recordings and oscillation detection in rats. Shown are simultaneously recorded signals from the motor cortex and hippocampus during natural sleep, illustrating detected slow oscillations, spindles, and sharp-wave ripples. These traces are the case of co-occurrence of sleep oscillations across brain regions.

Memory consolidation through sleep oscillations

During sleep, slow-oscillations and spindles are predominantly observed in the cerebral cortex, while sharp-wave ripples are evident in the hippocampus. These sleep-related oscillations regulate interactions between the hippocampus, cerebral cortex, and other brain regions to facilitate memory reactivation, and induce changes in neural representations and synaptic plasticity (23, 24). Notably, slow-oscillations and spindles increase following learning, and exhibit a significant correlation with memory reactivation (25–29).

Spindles are typically more frequently generated during the up-states of slow-oscillations (Fig. 1D), while a study suggests that neuronal spikes during the down-states also play a critical role in memory consolidation (1, 30). Research has shown that learning and memory are enhanced by artificially inducing spindles during the up-states of slow-oscillations (31, 32). Conversely, weakened synaptic connections in the cortex are associated with slow-oscillations that occur without spindles. Moreover, disruptions in the coordination between slow-oscillations and spindles have been linked to impaired learning and memory performance (33, 34).

Similarly, sharp-wave ripples generated in the hippocampal CA1 region play a crucial role in memory processing (35). Following learning, the coupling among slow-oscillations, spindles, and sharp-wave ripples increases; coupling refers to the precise temporal alignment of distinct sleep oscillations (Fig. 1D). This enhanced coupling has been associated with improved behavioral performance during memory recall tasks (36). An experiment where precise electrical stimulation was used to strengthen the synchronization between hippocampal sharp-wave ripples and other sleep oscillations demonstrated a reorganization of neural representations in the prefrontal cortex. This led to enhanced prefrontal responsiveness to learned material, resulting in superior memory retention and learning outcomes (37).

Furthermore, the recent study has demonstrated that during long-term motor learning, a two-stage modification in cross-area coordination among the prefrontal cortex, motor cortex, and hippocampal CA1 region—a dynamic transition in sleep-dependent processing from hippocampus-dependent early memory acquisition to hippocampus-independent late-stage cortical stabilization—plays a fundamental role in stabilizing neural representations that are associated with motor learning, ultimately contributing to memory consolidation (33).

Interestingly, emerging evidence suggests that “slow waves” during NREM sleep can be divided into two distinct types with different spatial and temporal characteristics (32, 38, 39). One class, referred to as slow-oscillations, tends to be more global and exhibits larger amplitudes, whereas the other class—often termed delta-waves—is more local with smaller amplitudes (9, 38, 40). A recent study in rats systematically classified these two waveforms and demonstrated that slow-oscillations and delta-waves have dissociable and even competing roles in motor memory processing during NREM sleep (32). Using

closed-loop optogenetic manipulation, the study revealed that slow-oscillations promote memory consolidation, while delta-waves are associated with forgetting.

Strikingly, these waves often occur in structured temporal sequences—such as a slow-oscillation followed by a cluster of delta-waves—suggesting that they may function as coordinated units of memory processing. This temporal arrangement may support an efficient balance between synaptic strengthening and weakening within each NREM sleep epoch, allowing for memory optimization without overwhelming network capacity. Future studies are needed to further characterize these functional units and their contribution to long-term memory stability and plasticity.

In contrast to the prominent slow-oscillations, spindles, and sharp-wave ripples of NREM sleep, theta oscillations (5–8 Hz), which dominate REM sleep, play a distinct yet critical role in memory consolidation. Theta activity is generated by interactions between the medial septum and hippocampus and is associated with enhanced synaptic plasticity, emotional memory processing, and the integration of newly acquired information into existing networks (41, 42). Studies have shown that hippocampal theta rhythms during REM sleep facilitate long-term stabilization of memory traces, particularly through phase-specific coordination with cortical activity and modulation of plasticity-related gene expression (43, 44). In both rodents and humans, theta oscillations during REM sleep have been linked to abstraction, associative learning, and the generalization of learned material—functions that complement the precise reactivation mechanisms of NREM sleep (45, 46). Thus, theta oscillations appear to support the integration and transformation of memories, forming the latter stage of a coordinated two-phase process in which NREM oscillations stabilize newly encoded content and REM theta refines and integrates them into broader knowledge networks.

In summary, extensive research highlights that during NREM sleep, the precise coordination and temporal coupling among slow-oscillations, spindles, and sharp-wave ripples play crucial roles in memory consolidation by facilitating memory reactivation and promoting neural network reorganization (Fig. 2A). Complementing these NREM-driven processes, theta oscillations during REM sleep appear to support the integration, abstraction, and emotional tagging of memories, aiding their incorporation into broader semantic frameworks. Together, these stage-specific oscillations form a dynamic, two-phase consolidation process in which NREM sleep stabilizes memory traces and REM sleep refines and transforms them for long-term storage and adaptive use.

Neuromodulators and memory consolidation during sleep

Each sleep stage is accompanied by distinct patterns of activity in neuromodulators, which play an essential role in memory consolidation and synaptic plasticity (47). This mini-review focuses on norepinephrine (NE) and dopamine (DA)—neuromodulators that are now being increasingly studied using

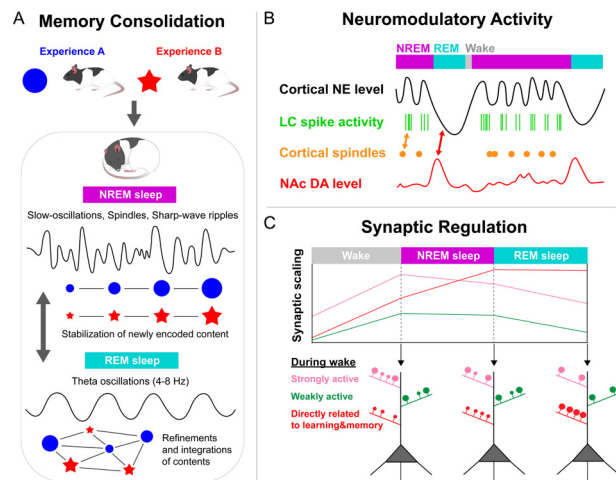


Fig. 2. Memory consolidation mechanisms across sleep stages. (A) Distinct roles of NREM and REM sleep in memory consolidation. During NREM sleep, the precise temporal coordination among slow oscillations, spindles, and sharp-wave ripples facilitates memory reactivation and neural network reorganization, supporting the stabilization of newly acquired memories. In contrast, REM sleep is dominated by theta oscillations, which contribute to the integration, abstraction, and emotional tagging of memories, enabling their incorporation into broader semantic frameworks. (B) Representative traces illustrate cortical NE fluctuations and DA dynamics across wakefulness, NREM sleep, and REM sleep in a mouse expressing a fluorescent NE sensor. During NREM sleep, NE levels oscillate slowly (~ 0.02 Hz), generating alternating phases of high and low spindle density. These fluctuations are temporally linked to bursts of *locus coeruleus* (LC) neuron activity, which coincide with the termination of individual spindles. Sustained low NE levels during NREM sleep are hypothesized to create a permissive state for synaptic consolidation (orange arrow). In contrast, REM sleep is marked by a sharp and sustained reduction in NE levels, accompanied by a surge in DA concentration in the *nucleus accumbens* (NAc), particularly during the NREM-to-REM transition (red arrow). This dopaminergic increase may signal a functional shift from stabilization to integration, supporting the abstraction and emotional tagging of memories. (C) Homeostatic synaptic downscaling and reorganization during sleep. Synaptic strength increases during wakefulness through information encoding. Sleep promotes synaptic downscaling, but the mechanisms differ by stage: NREM sleep selectively downscals highly active neurons, while REM sleep induces broader network-wide synaptic weakening. Memory-related synaptic assemblies are reactivated during NREM sleep, preserving their strength and potentially enhancing connectivity. The balance between global downscaling and memory-specific upscaling supports synaptic homeostasis and memory reorganization.

emerging *in vivo* optical monitoring techniques (e.g., fiberphotometry using sensors of GRAB and dLight)—they regulate neural dynamics during sleep, shaping the stabilization and reorganization of newly acquired memories (48).

Norepinephrine: Norepinephrine (NE) is primarily released from the *locus coeruleus* (LC), a brainstem structure that during wakefulness plays a central role in arousal regulation and attention (49, 50). However, during sleep, norepinephrine

levels fluctuate significantly, contributing to the state-dependent modulation of neural excitability and memory consolidation (51). During NREM sleep, norepinephrine levels oscillate at a slow frequency of 0.02 Hz, while during REM sleep, norepinephrine activity reaches its lowest levels (Fig. 2B) (52). These fluctuations are functionally significant, as studies have shown that norepinephrine signaling directly influences the frequency and density of spindles—key oscillations that are linked to memory consolidation (53). Specifically, when neurons in the LC fire, more spindles are generated to enhance the reactivation of learned experiences during sleep (Fig. 2B) (51, 53).

Furthermore, neural manipulations of norepinephrine levels provide insight into its role in memory stability. Suppressing norepinephrine activity too early during sleep disrupts the coupling between slow-oscillations and spindles, which is necessary for the cortical storage of declarative memories (54–56). On the other hand, artificially increasing norepinephrine levels during NREM sleep interferes with hippocampal–cortical communication, impairing the ability to consolidate new information (57). These findings suggest that the precise regulation of norepinephrine release during sleep is critical to optimize memory retention, while preventing excessive synaptic strengthening that could lead to interference between overlapping memory traces.

Dopamine: Like norepinephrine, dopamine (DA) plays a significant role in regulating memory consolidation, synaptic plasticity, and sleep-stage transitions (58–60). Throughout the sleep cycle, dopamine levels exhibit characteristic fluctuations (Fig. 2B), with a sharp increase in concentration marking the transition from NREM to REM sleep in the *nucleus accumbens* (NAc) and the *basolateral amygdala* (BLA) (59, 61). This increase is thought to signal a critical shift in memory processing, transitioning from stabilization and replay during NREM sleep, to integration and abstraction in REM sleep.

Beyond its role in sleep regulation, dopamine is a crucial modulator of memory salience and reinforcement learning (62). The role of the ventral tegmental area (VTA), a primary dopaminergic center, is key in linking reward, novelty, and memory formation (63). During wakefulness, VTA dopamine activity influences hippocampal reactivation during sleep, and may regulate the frequency of hippocampal sharp-wave ripples (64). Additionally, VTA dopamine neurons and dopamine release in the striatum exhibit delta-range fluctuations (65, 66). These findings indicate that dopamine release during NREM sleep aligns with memory-related oscillatory patterns, and helps selectively strengthen memories.

Norepinephrine and dopamine activity during sleep and the optimization of memory storage: Recent findings show that during sleep, dopamine and norepinephrine interact dynamically to regulate distinct phases of memory consolidation (67, 68). While LC-driven norepinephrine oscillations support memory consolidation through spindle clustering and synchronization, VTA dopamine activity aligns with memory-related oscillations, aiding prioritization and reinforcement

(47, 53). Additionally, the interaction of these oscillations with hippocampal-cortical oscillations may suggest that neuromodulation is a crucial factor in orchestrating the precise neural timing required for memory transfer and integration.

Acetylcholine and serotonin: Acetylcholine (ACh) exhibits stage-dependent activity, with the highest levels observed during REM sleep and the lowest during NREM sleep (69). Optogenetic activation of cholinergic neurons in the basal forebrain increases high-frequency cortical activity while suppressing slow waves, suggesting that a reduction in cholinergic tone is a prerequisite for NREM sleep (70). Although direct evidence remains limited, low ACh levels during NREM are thought to create a permissive environment for hippocampal-driven memory reactivation and redistribution, facilitating systems consolidation across distributed cortical networks. Similarly, serotonergic neurons show state-dependent discharge patterns, with high firing rates during wakefulness, reduced activity during NREM sleep, and near silence during REM sleep—closely resembling the pattern observed in LC neurons (71). Serotonin has also been associated with diminished sensory responsiveness and muscle atonia during sleep (72). However, the precise role of serotonin in regulating arousal and sleep-dependent memory processes remains incompletely understood. Due to the challenges of real-time *in vivo* monitoring, understanding how acetylcholine and serotonin interact with other neuromodulatory systems to influence sleep-dependent memory consolidation remains an important yet difficult task.

In summary, neuromodulatory activities, particularly those that involve norepinephrine and dopamine, play critical roles in coordinating transitions in sleep-stage, modulating memory reactivation, and selectively enhancing specific memories. The intricate balance of these neuromodulators ensures that important experiences are effectively consolidated, while preventing unnecessary memory interference. Understanding these mechanisms provides valuable insights into sleep-related learning processes, and may offer implications for interventions that target memory enhancement and cognitive function.

Synaptic regulation and memory consolidation during sleep

The capacity of the brain to store and manage information is thought to depend on synaptic scaling, a process that adjusts the strength of synaptic connections between neurons that encode memories (73, 74). While synaptic potentiation is crucial to learning and memory formation, excessive strengthening across the brain can lead to saturation, reducing the capacity for new information storage. To prevent this, sleep facilitates the selective weakening or elimination of synapses, a process now recognized as a fundamental mechanism of memory optimization and neural homeostasis (73).

Global synaptic downscaling during sleep: A widely accepted model to explain synaptic regulation during sleep is the Synaptic Homeostasis Hypothesis (SHY), proposed by Tononi and Cirelli (74–77). This model holds that during wakefulness, global synaptic potentiation occurs, as new experi-

ences and learning strengthen synaptic connections throughout the brain. This widespread strengthening, although beneficial in the short term, increases metabolic demand, causes greater synaptic noise, and reduces signal-to-noise ratio, which negative effects could ultimately impair neural efficiency. According to SHY, NREM sleep plays a critical role in resetting synaptic strength, selectively downscaling weaker synapses, while preserving the most relevant connections.

Many studies have demonstrated that during sleep, the contact area between cortical axon terminals and dendritic spines is globally reduced, while synaptic AMPA receptor levels also decrease to result in a measurable reduction in synaptic strength (74, 75, 78, 79). This downscaling process is thought to serve a homeostatic function to ensure that memory storage remains balanced and adaptable, rather than continuously accumulating, leading to saturation. Furthermore, the controlled reduction of synaptic weights during sleep enhances the brain's capacity to integrate new information the following day, facilitating continued learning and plasticity.

Synaptic plasticity during sleep balances weakening and reinforcement

While sleep is known to weaken synapses, recent research suggests it also plays a role in synaptic preservation and formation (80). Studies utilizing two-photon microscopy suggest that rather than indiscriminately weakening synapses, NREM sleep may actively preserve and promote the formation of new synaptic connections, particularly in the context of motor learning and skill acquisition (81). This dual mechanism of synaptic downscaling and targeted reinforcement aligns with behavioral findings that show that even in the absence of further practice, sleep enhances later motor skill learning (82, 83). Thus, procedural memory consolidation may depend on a selective balance between synaptic pruning and stabilization to reinforce relevant connections, while weakening unnecessary ones.

Synaptic refinement during REM sleep: In addition to its role in synaptic restructuring, REM sleep has been implicated in synaptic pruning and network reorganization. Studies have reported that following REM sleep, neuronal firing rates in both the hippocampus and cortex decrease, suggesting a role in neural refinement and optimization (43, 84). This supports the hypothesis that REM sleep fine-tunes neural circuits by weakening unnecessary synapses, preventing memory overload, and ensuring efficient processing. Furthermore, two-photon imaging studies have demonstrated that REM sleep does not just weaken synapses, but also plays an active role in selecting which synapses to maintain or remove (85). This selective process stabilizes only the most behaviorally relevant connections, while systematically pruning redundant or weakly activated synapses. Such targeted synaptic refinement may explain why REM sleep is particularly associated with emotional memory processing (86), and building unlearned inferential knowledge for problem-solving (87), as it helps optimize cortical networks for flexible and efficient recall.

Synaptic remodeling - strengthening and refining memory:

In addition to the large-scale reorganization of memory traces, sleep plays a critical role in the fine-tuning of synaptic connections (Fig. 2C). During the synaptic remodeling phase of memory consolidation, newly acquired memories undergo synaptic strengthening, whereas less relevant or redundant synaptic connections are selectively weakened (1, 73). This dual process preserves valuable experiences while preventing excessive synaptic potentiation that could interfere with future learning.

The relationship between systems consolidation and synaptic refinement remains incompletely understood, partly due to technical limitations in real-time monitoring of synaptic changes during sleep across distributed brain regions. Systems consolidation involves large-scale hippocampal-cortical interactions (1, 2), while synaptic refinement refers to local strengthening or weakening of connections that support memory precision and efficiency (76). Evidence suggests that refinement occurs in both the hippocampus and the cortex, with NREM and REM sleep contributing to synaptic potentiation and pruning, respectively (81, 85). However, how these local synaptic changes align temporally and anatomically with systems-level reorganization is still unclear, and future studies are needed to clarify their temporal and anatomical coordination.

Memory reactivation and its role in systems consolidation

Neural reactivation and the role of sleep oscillations in systems consolidation: Memory reactivation, a key mechanism of systems consolidation, is dominantly observed during NREM sleep, when newly encoded neural activity patterns are replayed in the hippocampus and cortex (19). This reactivation facilitates the transfer of memory representations from hippocampal-dependent storage to distributed cortical networks to enhance long-term retention and generalization. The precise coordination of multiple sleep oscillations, including slow-oscillations, spindles, and sharp-wave ripples, primarily drives sleep-dependent systems consolidation. These oscillations interact dynamically, with slow-oscillation up-states creating a window for spindles and hippocampal ripples, which synchronize neural reactivation in the hippocampus and cortex. The coupling of these oscillations has been strongly linked to successful memory consolidation, optimizing hippocampal – cortical communication, while ensuring the long-term stabilization of memory traces.

Neuromodulatory regulation of memory reactivation during sleep: While the role of neural oscillations in systems consolidation has become well-established, the influence of neuromodulatory activities on memory reactivation remains under investigation. Emerging evidence suggests that stage-specific fluctuations in neuromodulator levels, such as norepinephrine, dopamine, acetylcholine, and GABA, may play a crucial role in shaping sleep-dependent memory processing. For example, the suppression of norepinephrine during REM sleep has been proposed to facilitate synaptic remodeling (53), while dopaminergic activity during NREM sleep may help prioritize salient

memories for consolidation (48, 58). However, further research needs to determine the precise interplay between neuromodulatory changes and oscillation-driven memory replay across different sleep stages.

Interactions between NREM and REM sleep

Memory reactivation is a dynamic process that occurs across sleep, relying on the complementary roles of non-rapid eye movement (NREM) and rapid eye movement (REM) sleep. Rather than serving as isolated stages, NREM and REM interact in a temporally coordinated manner to support systems-level consolidation.

Complementary roles in memory reactivations: In rodents, memory reactivations have been predominantly observed during NREM sleep and periods of quiet wakefulness. As described above, these reactivations are closely linked to characteristic sleep oscillations—such as slow oscillations, spindles, and sharp-wave ripples (19, 88), which primarily occur during early sleep following learning. In contrast, only a limited number of studies have reported memory reactivation during REM sleep (89, 90). For instance, one study in rodents demonstrated reactivation of neural activity related to a four-trial sequence in a circular track task, modulated by sub-cortical theta rhythms (90).

Evidence also suggests that external cues during each stage of sleep may influence the type of reactivation. Several studies have shown that auditory stimuli presented during NREM sleep enhance hippocampal – cortical communication (91), whereas stimuli delivered during REM sleep preferentially promote corticocortical reactivations (92). The subsequent transfer of information to cortical networks during REM sleep may contribute to memory transformation processes, such as generalization, the formation of associative links, and integration with semantically related knowledge (93, 94). For example, in a rodent maze task, REM sleep was more likely to support the predictive association of familiar spatial memories linked to rewards (87, 90). Although still limited, current evidence suggests that the timescale and content of neural reactivations may differ depending on the sleep stage and the timing within the sleep cycle.

Complementary roles in synaptic regulation: Recent findings also indicate that NREM and REM sleep play distinct but complementary roles in synaptic reorganization (67, 72) (Fig. 2C). NREM sleep primarily facilitates synaptic potentiation, reinforcing neural connections involved in memory encoding and storage. In contrast, REM sleep appears to promote synaptic pruning, selectively weakening or eliminating less relevant or redundant connections to refine memory networks (60, 63). This interplay between strengthening and weakening of synapses supports both memory retention and cognitive flexibility.

Together, these findings suggest that NREM and REM sleep jointly support memory consolidation through both reactivation and synaptic remodeling. While NREM sleep strengthens key neural representations and stabilizes memory traces, REM

Table 1. List of featured recent studies cited in the current review

Category	Reference	Key finding
Memory consolidation	Lemke SM et al. (2021)	Motor-striatal coupling increases with skill learning during sleep
Memory consolidation	Kim J et al. (2022)	Sleep-dependent GABA, slow waves, and spindles restore memory after stroke
Memory consolidation	Senzai Y and Scanziani M (2022)	REM eye movements signal underlying cognitive activity
Memory consolidation	Sawangjit A et al. (2022)	Hippocampus is involved in tasks previously considered hippocampus-independent
Memory consolidation	Yang G et al. (2022)	REM-targeted cues promote memory integration
Memory consolidation	Maingret N et al. (2022)	Hippocampo-cortical coupling mediates sleep memory consolidation
Memory consolidation	Kim J et al. (2023)	Motor learning involves dynamic cortical-hippocampal coupling
Memory consolidation	Liu C, Todorova R et al. (2023)	Hippocampal codes enable associative and predictive memory guidance
Memory consolidation	Brodt S et al. (2023)	Sleep: a brain-state serving systems memory consolidation
Memory consolidation	Karaba LA et al. (2024)	Hippocampal circuit balances memory reactivation during sleep
Memory consolidation	Darevsky D et al. (2024)	Spindle train supports persistent reactivation events
Memory consolidation	Swanson RA et al. (2025)	Hippocampal ripples and cortical oscillations interact bidirectionally
Neuromodulatory activity	Osorio-Forero A et al. (2021)	NREM substates are regulated by noradrenergic circuits
Neuromodulatory activity	Kjaerby C et al. (2022)	Adrenergic tone dynamically shapes memory-related network activity during sleep
Neuromodulatory activity	Hasegawa M et al. (2022)	Rapid eye movement sleep is initiated by dopamine signaling
Neuromodulatory activity	Duran C et al. (2023)	Norepinephrine dynamics during NREM to support memory consolidation
Neuromodulatory activity	Sulaman BA et al. (2024)	SFN symposium: functions of neuromodulation during sleep
Neuromodulatory activity	Toth AB et al. (2025)	Dopamine release in the NAc regulates the timing of REM sleep
Synaptic regulation	Weiss JT and Donlea JM (2021)	Review: sleep loss impacts synaptic plasticity
Synaptic regulation	Koukaroudi et al. (2024)	Sleep maintains excitatory synapse diversity
Synaptic regulation	Sawada T, Iino Y et al. (2024)	Sleep-dependent synaptic regulation through synaptic chemogenetics
Dream	Malinowski JE and Horton CL (2021)	Dream content reflects sleep cognition and predicts memory performance
Dream	Picard-Deland C et al. (2023)	Demonstrated real-time communication with dreamers during NREM sleep
Dream	Horowitz AH, Esfahany K et al. (2023)	Targeted dream incubation at sleep onset enhances creativity after sleep
Dream	Lacaux C et al. (2024)	Identified sleep onset as a critical state for boosting creative insight

This table provides a curated list of recent studies (2021–2025) cited in the current review, organized by thematic categories: Memory consolidation, Synaptic regulation, Neuromodulatory activity, and Dream. For each study, we include the full citation, publication year, and a brief summary of the key finding or contribution relevant to sleep-dependent memory processing. The categorization reflects the primary focus of each study as interpreted in the context of this review. Studies were selected based on their empirical or theoretical contribution to understanding systems-level consolidation, local synaptic plasticity, neuromodulator-guided dynamics, or the cognitive and phenomenological aspects of dreaming.

sleep contributes to the optimization and integration of these memories by removing nonessential connections and promoting network refinement. This balanced mechanism enhances learning capacity, cognitive efficiency, and the long-term stability of stored information. A deeper understanding of these complementary processes may offer valuable insights into sleep-dependent learning, neuroplasticity, and interventions for memory-related disorders.

Clinical implications of systems consolidation

Understanding the mechanisms of systems consolidation during sleep has growing clinical relevance, particularly for neuropsychiatric and neurodegenerative disorders characterized by memory impairments. Disruptions in sleep architecture and oscillatory coordination are commonly observed in conditions such as Alzheimer's disease, Parkinson's disease, schizophrenia,

and depression (95, 96), where deficits in slow-oscillations, spindles, or ripple coupling have been linked to impaired memory consolidation (32). In addition, abnormal neuromodulatory dynamics—such as altered norepinephrine or dopamine signaling—may further interfere with memory reactivation and storage (50). These findings highlight the potential of targeting sleep physiology through pharmacological or neuromodulatory interventions to enhance memory processing.

Closed-loop stimulation techniques that modulate specific sleep oscillations in real time offer promising avenues for restoring memory function (97, 98). Notably, a recent study demonstrated that neuromodulation aimed at enhancing up-states during sleep can promote recovery following stroke (99). It is conceivable that closed-loop neuromodulatory approaches—designed to suppress delta-waves (associated with forgetting) or enhance slow-oscillations (associated with conso-

lidation)—could provide substantial long-term benefits in preventing cognitive decline or improving treatment outcomes. Future research should aim to translate mechanistic insights into sleep-dependent memory consolidation into effective therapeutic strategies for clinical populations.

Limitations of current systems consolidation models

Despite significant advances, current models of systems consolidation remain incomplete and, in some cases, conflicting. The standard consolidation theory posits a time-limited role for the hippocampus, suggesting that memories gradually become hippocampus-independent as they are transferred to the neocortex (1, 2). However, considerable evidence from both animal and human studies challenges this view, showing persistent hippocampal involvement in the retrieval of remote episodic memories (100, 101). In contrast, the multiple trace theory and its variants propose that detailed episodic memories always depend on the hippocampus (102, 103), yet these frameworks do not fully explain the dynamic reorganization and apparent degradation of hippocampal traces over time.

Recent findings suggest that hippocampal memory representations are inherently unstable—subject to synaptic turnover, neurogenesis, and random remapping—raising questions about how stable memory content is retrieved after the original trace has faded (104, 105). The scene construction theory offers one resolution, proposing that the hippocampus reconstructs remote experiences from neocortical elements rather than retrieving a preserved trace (101). While promising, this framework still requires empirical clarification, particularly regarding how cortical and hippocampal interactions enable reconstruction. Additionally, existing models often oversimplify the contributions of sleep-specific oscillations and neuromodulatory dynamics in guiding which memories are consolidated, transformed, or pruned (32, 37). Together, these limitations underscore the need for integrative theories that account for both the transience and reconstructive capacity of hippocampal-cortical memory, the temporal structure of sleep, and the selective pressures on memory updating over time.

Future perspectives for sleep-dependent memory consolidation

Sleep-dependent memory consolidation involves multiple interconnected processes. Sleep oscillations act as key temporal organizers, orchestrating hippocampal–cortical interactions to facilitate systems memory transfer (1, 2, 19). Simultaneously, neuromodulatory regulation fine-tunes these interactions, potentially influencing which memories are prioritized for long-term storage (47, 53). Finally, synaptic remodeling during sleep stabilizes newly formed connections while pruning unnecessary ones, to optimize the brain's capacity for learning and adaptation (76, 77).

Several decades of sleep research have established the critical role of two fundamental sleep stages—NREM and REM sleep—in memory consolidation. In parallel, a few studies

have suggested that dreaming (i.e., any conscious experience during sleep) may reflect the manifestation of memory reactivation processes occurring during sleep (106). Dream experiences vary markedly across sleep stages. While dreams are more frequently recalled and emotionally vivid during REM sleep, they also occur during NREM sleep, particularly at sleep onset (N1) (107–110). NREM dreams are typically shorter, more realistic, and rich in episodic content, whereas REM dreams are longer, more narrative-driven, emotionally intense, and enriched with semantic associations (111). These distinctions likely arise from differences in the neurophysiological environment—such as neuromodulatory dynamics and cortical activation—that define each sleep stage and influence both the generation and recall of dreams.

The temporal source of dream content also varies. Recent experiences, or “day residues,” frequently appear in early-night and N1 dreams, while late-night REM dreams are more likely to incorporate remote memories. A “dream-lag effect” has been documented, in which events that occurred 5–7 days earlier resurface during REM dreams (112). Moreover, distant memories may be recruited into dreams through associative links with recent experiences, potentially facilitating the integration of novel information into existing memory networks. Although the precise neural mechanisms underlying dreaming remain debated, future research employing innovative methods will be crucial for uncovering the complex interactions between memory consolidation and dream generation, as well as for advancing our understanding of the functional significance of dreaming.

Notwithstanding that significant progress has been made in understanding the neural and molecular mechanisms underlying sleep-dependent memory consolidation (Table 1), many questions remain. Future research should seek to further clarify the precise role of neuromodulators in sleep oscillation dynamics, how NREM and REM sleep optimize memory storage, and the impact of sleep-dependent synaptic reorganization on cognitive function. More profound understanding of these mechanisms will advance our knowledge of memory processing, while also potentially providing insights into therapeutic interventions for sleep disorders and memory-related impairments.

CONCLUDING REMARKS

Systems-level memory consolidation during sleep involves the gradual reorganization of neural connections, particularly between the hippocampus and neocortex (1, 2). This process is orchestrated by coordinated sleep oscillations, neuromodulatory dynamics, and synaptic remodeling, with NREM and REM sleep contributing complementary functions—NREM supporting memory reactivation and stabilization, and REM promoting synaptic refinement and integration.

Recent findings suggest that dreaming may reflect aspects of these consolidation processes, particularly through the integration of recent and remote memories during sleep. While

the precise mechanisms remain to be clarified, emerging evidence links dream content to stage-specific memory reactivation and transformation. Future research should further explore how conscious experiences during sleep relate to the underlying neural dynamics of memory consolidation.

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CONFLICTS OF INTEREST

The authors have no conflicting interests.

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